Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/06/2008
PI Name:	Clark, Noel A. Ph.D.		
Project Title:	Structure and Dynamics of Freely Suspended Liquid Crystals		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	noel.clark@colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	303-941-1008
Organization Name:	University of Colorado		
PI Address 1:	Physics Department, CB390		
PI Address 2:	2000 Colorado Ave		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0001	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Griffin, DeVon	**Contact Phone:**	216-433-8109
Contact Email:	DeVon.Griffin@grc.nasa.gov , DeVon.W.Griffin@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: funds reinstated until 11/30/2007, per J. Cohen (HQ) 8/2006. [Note - end date changed from 11/30/2003 grant renewed for 4 yrs/new NAG#]		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Maclennan, Joseph (University of Colorado) Glaser, Matthew (University of Colorado)		
Grant/Contract No.:	NNC04GA50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

We pursue ground-based studies of freely suspended liquid crystal films with an eye toward µg experiments, as well as atomistic computer simulation of liquid crystal phases, probing the molecular origins of observed film behavior. Ground-based studies focus on tethered smectic bubbles on which islands (isolated pancake-shaped thicker regions) can be generated and observed. The islands form a novel interacting particle array confined to the two-dimensional surface of the bubble. We observe islands (using reflected light microscopy) and manipulate them (using optical tweezers) and carry out computer simulation of the interaction of islands on contact, in an effort to understand their stability against aggregation.

Our work is directed toward developing a deeper understanding of condensed phases under conditions of reduced dimensionality. Films are locally optically anisotropic so that the pattern of molecular orientation within the film plane can be effectively studied by depolarized reflected light microscopy are also being studied. In particular, topological defect motion can be observed and the simplest liquid crystal defect dynamical process, the attraction and annihilation of oppositely charged, unit-strength defects (vortex-antivortex pairs), probed. Experiments to date reveal significant shortcomings in our understanding of this process. Application of in-plane electric fields to tilted smectic films reveals a host of interesting field-induced structures and transitions, and leads to novel methods for probing film structure. For example, by comparing the response of even layer numbered and odd layer numbered films, basic information about the nature of the polar ordering (ferroelectric or antiferroelectric) of the smectic can be obtained. Our recent discovery of the spontaneous formation of chiral domains in fluid smectic phases of achiral banana-shaped molecules (the first observation of spontaneous chiral domain formation in a fluid since Pasteur's discovery in crystals 150 years ago), based on observation of free film response to applied electric field, will be pursued, as will the recent discovery of freely suspended filaments of smectics comprised of banana-shaped molecules.

Smectic liquid crystals are phases of rod shaped molecules organized into one dimensionally (1D) periodic arrays of layers, each layer being a two dimensional (2D) liquid between one and two molecular lengths thick. This inherent fluid-layer structure and the low vapor pressure of smectic liquid crystals enable the long term stabilization of freely suspended, single component, fluid films quantized in integral numbers of layers in thickness, down to a single molecular layer (30Å). These ultrathin freely suspended liquid crystal films are structures of fundamental interest in condensed matter and fluid physics. They are the thinnest known stable condensed phase fluid structures and have the largest surface-to-volume ratio of any stable fluid preparation, making them ideal for the study of the effects of reduced dimensionality on phase behavior and on fluctuation and interface phenomena. Their low vapor pressure and quantized thickness enable the effective use of microgravity to extend the study of basic capillary phenomena to ultrathin fluid films.

Ed. NOTE (5/29/14): Project continued with PI Noel Clark as "Structure and Dynamics of Freely Suspended Liquid Crystals--NNX07AE48G," with period of performance of 4/12/2007-4/11/2013. See that project for subsequent reporting.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Industry Affiliates:
Instec, Inc., Boulder, CO, Imaging Technology International, Boulder, CO Point Technologies, Inc. Boulder, CO Displaytech, Inc. Longmont, CO

Innovative Technologies Developed:

development of ultrathin bubbles, dispersion system of nanodroplet of liquid crystals, manipulation technique of nanodroplet of liquid crystals

Research Audience: physicists, chemist, engineers, materials scientist

Graduate Student Employment:

Nattaporn Chattham - Assistant Professor at Kanseisart University in Bangkok, Thailand Apichart Pattanaporkratana - Assistant Professor at Kanseisart University in Bangkok, Thailand

Research Impact on American/Earth Benefits: fundamental studies of fluid dynamics and self-organization in two dimensions

Task Progress:

Quadruple Interactions between Chiral Vortices of Opposite Handedness in the 2D XY Field of Freely-Suspended Racemic Smectic C Liquid Crystal Films:
In a few-layer thick freely suspended tilted smectic liquid crystal film, the local azimuthal orientation of the tilt (x,y) is a two-dimensional XY field. An island on the film (a circular region of greater thickness), presents a strong azimuthal boundary condi-tion at its edge that forces a strength +1 chiral vortex into XY field. Each such vortex is paired with a strength -1 vortex in the field, forming a chiral topological dipole. Chiral dipoles with the same handedness form polar chains with the dipoles pointing in the same direction and along the chain, whereas vortices of opposite handedness form quadrupolar structures. XY elasticity predicts that islands should touch in such quadrupole but in fact are found to exhibit a minimum in their interactive energy at finite separation. We suggest that the reason for this bias is that the spontaneous polarization of the chiral SmC* phase would prefer to either point toward or away from the liquid crystal-air interface at the edges of the islands, inducing vortices of only one handedness in the islands.

Numerical Modeling of the Elastic Interactions of Vortices and Islands in the Quadrupolar Structure:

We have modeled the elastic interactions of vortices and islands in the quadrupolar structure numerically using a finite element method with adaptive meshing. The energy potential is qualitatively similar to that of dipolar chains, being attractive at long distances, repulsive at small distances, and minimal at an equilibrium separation D0 which depends on the elastic anisotropy. As kappa increases, the chiral islands almost touch each other.

Ed. NOTE (5/29/14): Project continued with PI Noel Clark as "Structure and Dynamics of Freely Suspended Liquid Crystals--NNX07AE48G," with period of performance of 4/12/2007-4/11/2013. See that project for subsequent reporting.

Bibliography Type:	Description: (Last Updated: 12/04/2024)
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Interaction between structually chiral islands on freely-suspended smectic C films." March Meeting of the American Physical Society, Denver, CO, March 2007. Bulletin of the American Physical Society 2007 Mar;52. , Mar-2007
Abstracts for Journals and Proceedings	Chenhui Zhu A, Pattanaporkratana A, Maclennan JE, Clark NA. "Topological defect structure and annihilation in high-polarization freely suspended films." March Meeting of the American Physical Society, Denver, CO, March 2007. Bulletin of the American Physical Society 2007 Mar;52. , Mar-2007
Abstracts for Journals and Proceedings	Chenhui Zhu, Muzny C, Tewary A, Link D, Fritz A, Coleman D, Maclennan JE, Clark NA. "Annihilation of point defects in smectic C liquid crystal films." March Meeting of the American Physical Society, Denver, CO, March 2007. Bulletin of the American Physical Society 2007 Mar;52. , Mar-2007
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Effect of polarization on the interactions of smectic C* islands on freely-suspended films." March Meeting of the American Physical Society, Denver, CO, March 2007. Bulletin of the American Physical Society 2007 Mar;52. , Mar-2007
Abstracts for Journals and Proceedings	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Interactions between chiral vortices in the 2d xy field of freely-suspended smectic C liquid crystal films." 11th International Conference on Ferroelectric Liquid Crystals, Sapporo, JAPAN, September 2007. Proceedings of the 11th International Conference on Ferroelectric Liquid Crystals, 2007. , Sep-2007
Abstracts for Journals and Proceedings	Nguyen D, Park CS, Maclennan JE, Glaser M, Clark NA. "Coarsensing of two-dimensional islands in freely suspended smectic A films." March Meeting of the American Physical Society, New Orleans, LA, March 2008. Bulletin of the American Physical Society 2008 Mar;53. , Mar-2008
Abstracts for Journals and Proceedings	Nguyen D, Park CS, Maclennan JE, Glaser M, Clark NA. "Coarsensing of two-dimensional islands in freely suspended smectic A films and bubbles." 21st International Liquid Crystal Conference, Jeju Island, Korea, June 30 - July 4, 2008. Proceedings of 22st International Liquid Crystal Conference, June 2008. , Jun-2008
Articles in Peer-reviewed Journals	Nakata M, Shao RF, Maclennan JE, Weissflog W, Clark NA. "Electric-field-induced chirality flipping in smectic liquid crystals: the role of anisotropic viscosity." Physical Review Letters. 2006 Feb 17;96(6):067802. PMID: 16606050 , Feb-2006
Articles in Peer-reviewed Journals	Pattanaporkratana A, Park CS, Maclennan JE, Clark NA. "Interaction between chiral vortices in the 2d xy field of freely-suspended smectic C liquid crystal films." Physical Review Letters. Submitted, June 2008. , Jun-2008
Articles in Peer-reviewed Journals	Lansac Y, Clark NA, Glaser MA. "Nanophase segregation and frustration: spontaneous chirality and polarization." Physical Review Letters. Submitted, May 2008. , May-2008
Articles in Peer-reviewed Journals	Chattham N, Korblova E, Shao R, Walba DM, Maclennan JE, Clark NA. "Triclinic fluid order." Physical Review Letters. 2010 Feb;104(6):067801. http://dx.doi.org/10.1103/PhysRevLett.104.067801 , Feb-2010
Dissertations and Theses	Pattanaporkratana A. "Textures and Interactions between Vortices in the 2D XY Field of Freely Suspended SmC and SmC* Liquid Crystal Films." Dissertation, Department of Physics, University of Colorado, June 2007. , Jun-2007